

Brahadeesh Sankarnarayanan

Assistant Professor (Grade II), Department of Mathematics
Indian Institute of Technology Jodhpur, NH 62 Nagaur Road,
Karwar 342030, Jodhpur District, Rajasthan, India

brahadeesh@iitj.ac.in · +91 291 280 1469

[Website](#) · [Google Scholar](#) · [ORCID](#) · [MathSciNet](#) · [zbMATH Open](#)

RESEARCH INTERESTS

I work in extremal combinatorics and graph theory. My current interests are in spectral extremal graph theory, intersection graphs of paths, intersecting set systems, graph convexity, and majority coloring games. I have also worked on colorings and list colorings of graphs embedded in surfaces, and on the ranks of structured matrices arising from tournaments.

CURRENT PROJECTS

Majority coloring game on trees

With Saraswati Girish Nanoti (IISc Bangalore). Investigating optimal strategies and computational complexity for the two-player majority coloring game, particularly on trees.

Δ -convexity in graphs

Investigating convexity structures in graphs, particularly computing dimensional parameters such as the exchange number and Carathéodory number for Δ -convexity, and characterizing the extremal examples.

Brualdi–Hoffman-type extremal problems

With M. Rajesh Kannan (IIT Hyderabad). Investigating spectral extremal problems for F -free graphs on m edges, extending recent results for the case $F = K_3$ to $F = K_4$ and beyond.

Intersection graphs of paths

With Niranjana Balachandran (IIT Bombay), Shagnik Das (National Taiwan University), Umesh Shankar (IISc Bangalore). Investigating the path-representation number $\text{prn}(G)$ of a graph G : bounds in terms of other graph parameters, algorithms for computing optimal path representations, computational complexity of the parameter for different graph classes.

Fractional intersecting families

Continuing investigation of the Balachandran–Mathew–Mishra conjecture on the maximum size of fractional θ -intersecting families over $[n]$, building on [Balachandran–Das–Sankarnarayanan \(2025\)](#) and [Balachandran–Bhattacharya–Kher–Mathew–Sankarnarayanan \(2023\)](#).

EDUCATION

Ph.D. (Mathematics)

07/2018 – 08/2024

Indian Institute of Technology (IIT) Bombay

Thesis: “Some problems in combinatorics: Excursions in graph colorings and extremal set theory”

Advisor: Niranjana Balachandran

CPI: 8.89/10.0

BS–MS (Dual) Degree (Major in Mathematics)

07/2012 – 06/2017

Indian Institute of Science Education and Research (IISER) Bhopal

Thesis: “Automorphic forms and Tate’s thesis”

Advisor: Karam Deo Shankhadhar

CPI: 9.31/10.0

EMPLOYMENT

Assistant Professor (Grade II)

09/2025 – present

Department of Mathematics, IIT Jodhpur

Research Associate

05/2025 – 09/2025

Department of Mathematics, IIT Hyderabad

Project: “Spectral Theory of Signed Graphs”

Principal Investigator: M. Rajesh Kannan

Institute Post-Doctoral Fellow

08/2024 – 02/2025

Department of Mathematics, IIT Bombay

Mentor: Niranjana Balachandran

Project Assistant

Department of Mathematics, IISER Bhopal

Project: “Self-dual representations and signs”

Principal Investigator: Kumar Balasubramanian

09/2017 – 11/2017

PUBLICATIONS

REFEREED JOURNAL ARTICLES

Niranjana Balachandran, Brahadeesh Sankarnarayanan.

5-list coloring toroidal 6-regular triangulations in linear time.

Discrete Applied Mathematics **386** (2026), 75–103, [doi:10.1016/j.dam.2026.01.032](https://doi.org/10.1016/j.dam.2026.01.032),
[arXiv:2106.01634](https://arxiv.org/abs/2106.01634), [MR5028701](https://mrindex.org/5028701), [Zbl 08187875](https://zbmath.org/08187875).

Niranjana Balachandran, Shagnik Das, Brahadeesh Sankarnarayanan.

Bounded fractional intersecting families are linear in size.

The Electronic Journal of Combinatorics **32** (2025), no. 3, #P3.34, 10 pp., [doi:10.37236/12900](https://doi.org/10.37236/12900),
[arXiv:2402.14981](https://arxiv.org/abs/2402.14981), [MR4949605](https://mrindex.org/4949605), [Zbl 1572.05240](https://zbmath.org/1572.05240).

Niranjana Balachandran, Brahadeesh Sankarnarayanan.

Low-rank matrices, tournaments, and symmetric designs.

Linear Algebra and its Applications **694** (2024), 136–147, [doi:10.1016/j.laa.2024.04.006](https://doi.org/10.1016/j.laa.2024.04.006),
[arXiv:2401.14015](https://arxiv.org/abs/2401.14015), [MR4736676](https://mrindex.org/4736676), [Zbl 1539.05071](https://zbmath.org/1539.05071).

Niranjana Balachandran, Srimanta Bhattacharya, Brahadeesh Sankarnarayanan.

Almost full rank matrices arising from transitive tournaments.

Linear and Multilinear Algebra **72** (2024), no. 2, 153–161, [doi:10.1080/03081087.2022.2158168](https://doi.org/10.1080/03081087.2022.2158168),
[MR4703565](https://mrindex.org/4703565), [Zbl 1533.05147](https://zbmath.org/1533.05147).

Niranjana Balachandran, Srimanta Bhattacharya, Krishn Vishwas Kher, Rogers Mathew, Brahadeesh Sankarnarayanan.

On hierarchically closed fractional intersecting families.

The Electronic Journal of Combinatorics **30** (2023), no. 4, #P4.37, 17 pp., [doi:10.37236/11651](https://doi.org/10.37236/11651),
[arXiv:2211.02540](https://arxiv.org/abs/2211.02540), [MR4672585](https://mrindex.org/4672585), [Zbl 1533.05268](https://zbmath.org/1533.05268).

Niranjana Balachandran, Srimanta Bhattacharya, Brahadeesh Sankarnarayanan.

An ensemble of high-rank matrices arising from tournaments.

Linear Algebra and its Applications **658** (2023), 310–318, [doi:10.1016/j.laa.2022.11.004](https://doi.org/10.1016/j.laa.2022.11.004),
[arXiv:2108.10871](https://arxiv.org/abs/2108.10871), [MR4513895](https://mrindex.org/4513895), [Zbl 1506.15034](https://zbmath.org/1506.15034).

Brahadeesh Sankarnarayanan.

Note on 4-coloring 6-regular triangulations on the torus.

Annals of Combinatorics **26** (2022), no. 3, 559–569, [doi:10.1007/s00026-022-00573-8](https://doi.org/10.1007/s00026-022-00573-8),
[arXiv:2106.01037](https://arxiv.org/abs/2106.01037), [MR4473892](https://mrindex.org/4473892), [Zbl 1497.05098](https://zbmath.org/1497.05098).

Niranjana Balachandran, Brahadeesh Sankarnarayanan.

The choice number versus the chromatic number for graphs embeddable on orientable surfaces.

The Electronic Journal of Combinatorics **28** (2021), no. 4, #P4.50, 20 pp., [doi:10.37236/10263](https://doi.org/10.37236/10263),
[arXiv:2102.06993](https://arxiv.org/abs/2102.06993), [MR4394670](https://mrindex.org/4394670), [Zbl 1486.05101](https://zbmath.org/1486.05101).

Kumar Balasubramanian, K. S. Senthil Raani, Brahadeesh Sankarnarayanan.

Self-dual representations of $SL(2, F)$: an approach using the Iwahori–Hecke algebra.

Communications in Algebra **47** (2019), no. 10, 4210–4215, [doi:10.1080/00927872.2019.1581211](https://doi.org/10.1080/00927872.2019.1581211),
[MR3975999](https://mrindex.org/3975999), [Zbl 1432.20031](https://zbmath.org/1432.20031).

CONFERENCE PROCEEDINGS

Niranjana Balachandran, Brahadeesh Sankarnarayanan.

5-list coloring toroidal 6-regular triangulations in linear time. Extended abstract.

CALDAM 2023. In: Amitabha Bagchi, Rahul Muthu (eds.) *Algorithms and Discrete Applied Mathematics*, Lecture Notes in Computer Science, vol. 13947, pp. 134–146, Springer, Cham (2023),
[doi:10.1007/978-3-031-25211-2_10](https://doi.org/10.1007/978-3-031-25211-2_10), [MR4573050](https://mrindex.org/4573050), [Zbl 07728658](https://zbmath.org/07728658).

The full version appeared in Discrete Applied Mathematics **386** (2026), 75–103 (above).

MANUSCRIPTS IN PREPARATION

Niranjan Balachandran, Shagnik Das, Brahadeesh Sankarnarayanan, Umesh Shankar.
An extremal problem for vertex intersection graphs of paths.
2026, in preparation.

RESEARCH SUPERVISION

DOCTORAL STUDENTS

Vishnu Kumar Department of Mathematics, IIT Jodhpur	enrolled Autumn 2025–26
Yash Chawda Department of Mathematics, IIT Jodhpur	enrolled Spring 2025–26

EXECUTIVE M.TECH. PROJECTS

Rayipalli Prakash School of Artificial Intelligence and Data Science, IIT Jodhpur Project: “GNN-Based Automatic Calorie Identification from Food Images”	10/2025 – present
Md. Parvez Rahaman School of Artificial Intelligence and Data Science, IIT Jodhpur Project: “AI-Based Text Summarization and Tone Rewriting System Using BART and T5”	10/2025 – present

SUMMER INTERNS

Vaibhav Suvagiya SURAJ Programme, IIT Jodhpur Home institution: Sardar Vallabhbhai National Institute of Technology, Surat	Summer 2026
B. James Mejulan SURAJ Programme, IIT Jodhpur Home institution: Sardar Vallabhbhai National Institute of Technology, Surat	Summer 2026

FELLOWSHIPS, AWARDS & HONORS

Institute Post-Doctoral Fellowship IIT Bombay	08/2024 – 02/2025
National Board for Higher Mathematics (NBHM) Research Award Department of Atomic Energy, Government of India	08/2018 – 07/2023
Joint CSIR–UGC National Eligibility Test for Junior Research Fellowship CSIR – Human Resource Development Group; All-India Rank 20 in Mathematical Sciences	07/2017
Director’s Gold Medal IISER Bhopal Awarded for outstanding all-round achievement and leadership in the BS–MS graduating class	06/2017
Rashtrapati Bhavan In-Residence Programme for Scholars from IITs, IISERs, and IISc Invited by the President of India to represent IISER Bhopal	06/2016
Science Academies’ Summer Research Fellowship Indian Academy of Sciences, Indian National Science Academy, and National Academy of Sciences, India	05/2015 & 05/2016
World Science Conference, Israel Hebrew University of Jerusalem; member of a 15-person Indian contingent	08/2015
INSPIRE Scholarship for Higher Education Department of Science & Technology, Government of India	08/2012 – 06/2017

RESEARCH GRANTS

Research Initiation Grant (RIG) Indian Institute of Technology Jodhpur	03/2026 – present
--	-------------------

Amount: Rs. 4,00,000
Role: Principal Investigator

TALKS

TECHNICAL

- Competency-based Learning in Mathematics** 04/2026
Session at the STEM Teacher Training Program, organized (in association with Jodhpur Zila Parishad and Shiksha Vibhag) by the Centre for Education Technology, IIT Jodhpur
- Nonstandard analysis: a gentle introduction** 12/2025
Discussion Meeting on Number Theory and Combinatorics, Evolve Back, Coorg
- Bisection-closed families, tournaments, and symmetric designs** 05/2025
Geometry, Probability, and Algorithms, International Centre for Theoretical Sciences, Bengaluru
- Sunflowers and symmetric designs** 04/2025
Research Seminar, Department of Mathematics, IIT Jodhpur
- Sunflowers and symmetric designs** 11/2024
Research Associate Seminar, Theoretical Statistics and Mathematics Unit, Indian Statistical Institute, Bangalore Centre
- Some recent work on fractional intersecting families** 06/2024
Meru Annual Combinatorics Conference, Graphic Era Hill University, Bhimtal Campus
- Sunflowers, tournaments, and symmetric designs** 05/2024
Institute Post-Doctoral Fellowship Seminar, Department of Mathematics, IIT Bombay
- Some recent work on fractional intersecting families** 03/2024
CSE Research Symposium, IIT Palakkad

POPULAR AND EXPOSITORY

- A Combinatorial Gem of Maryam Mirzakhani: List Colorings of Planar Graphs** 05/2026
One Day Symposium Celebrating Women in Mathematics, IIT Jodhpur
- Counting the Infinite** 12/2025
National Mathematics Day Guest Lecture, Government College Makroniya Bujurg, Sagar, Madhya Pradesh (online)

EVENTS ORGANIZED

- Screening of *Secrets of the Surface: The Mathematical Vision of Maryam Mirzakhani*** 05/2026
Documentary film by George Csicsery; screening organized under the Mathematics Society as a follow-up to the One Day Symposium Celebrating Women in Mathematics
Department of Mathematics, IIT Jodhpur
- Career Opportunities Beyond Academia** 03/2026
One-day interaction session with industry mathematicians
Department of Mathematics, IIT Jodhpur
- An Introductory Workshop on the Lean Theorem Prover** 02/2026
Three-day workshop led by T. V. H. Prathamesh (Krea University);
co-organized with Dilpreet Kaur and Kriti Goel
Department of Mathematics, IIT Jodhpur

TEACHING EXPERIENCE

INSTRUCTOR: IIT JODHPUR

- MAL2040 Design and Analysis of Experiments** Spring 2025–26
49 students; overall instructor rating 3.70/5.0 (mean), median 4 across all dimensions

TEACHING ASSISTANT: IIT HYDERABAD

MA5020 Functional Analysis

Autumn 2025–26

TEACHING ASSISTANT: IIT BOMBAY

SI 419 Combinatorics

Autumn 2024–25, Autumn 2021–22

SI 416 Optimization

Spring 2021–22

MA 205 Complex Analysis

Autumn 2021–22, Autumn 2020–21, Autumn 2019–20

MA 207 Differential Equations II

Autumn 2020–21, Autumn 2019–20

MA 108 Differential Equations

Summer 2020–21, Spring 2019–20

MA 106 Linear Algebra

Spring 2020–21, Spring 2019–20

MA 111 Calculus II

Spring 2020–21

OTHER TEACHING AND MENTORING

Masterminds Program (Trainer)

2023–24, 2024–25

UAE IMO selection camp, Ministry of Education, UAE

Topics: combinatorics and number theory

Self-Assessment Team (Member)

09/2023 – 12/2023

Mathematics Training and Talent Search Programme (MTTS)

Online Foundation Course in Mathematics (Mentor)

08/2023 – 09/2023

Mathematics Training and Talent Search Programme (MTTS)

Online Short Course in Linear Algebra: Application View (Mentor)

07/2023

Mathematics Training and Talent Search Programme (MTTS)

Study-Help Room (Co-organizer)

2023

Department of Mathematics, IIT Bombay

Weekly drop-in math help for BS and MSc students

TA Skill Enhancement & Training (Master TA)

10/2022 – 11/2022

Parimal and Pramod Chaudhari Centre for Learning and Teaching, IIT Bombay

A Basic Course in Number Theory (TA)

07/2020 – 10/2020

NPTEL Online Certification course

PROFESSIONAL SERVICE

MEMBERSHIPS

Associate Fellow, Institute of Combinatorics and its Applications

05/2026 – present

Life Member, The Indian Mathematical Society

05/2026 – present

Life Member, Mathematics Teachers' Association (India)

05/2026 – present

Affiliate Member, American Mathematical Society

01/2026 – present

Advisory Board Member, Heliyon Mathematics (Cell Press)

09/2024 – 02/2025

REVIEWING

Indian Journal of Pure and Applied Mathematics

12/2025 – present

Acta Informatica

11/2025 – present

Proceedings – Mathematical Sciences (Indian Academy of Sciences)

04/2024 – present

Mathematical Reviews / MathSciNet

03/2023 – present

zbMATH Open

01/2022 – present

Conference on Algorithms and Discrete Applied Mathematics (CALDAM)

10/2020 – present

INSTITUTIONAL SERVICE

IIT JODHPUR

Faculty Advisor, Mathematics Society Board of Departmental Societies	09/2025 – present
Member, Departmental Undergraduate Committee (DUGC) Department of Mathematics	09/2025 – present
Member, Department-level committee for projection of new B.S. programme B.S. in Mathematics and Computing, Department of Mathematics	10/2025 – present

IIT BOMBAY

Coordinator, Students' Seminar Department of Mathematics	01/2023 – 05/2023
--	-------------------

IISER BHOPAL

President, Students Activity Council	02/2016 – 01/2017
Coordinator, Mathematics Club	08/2013 – 07/2015

CONFERENCES, WORKSHOPS & EXTERNAL COURSES ATTENDED

LeanLang for Verified Autonomy Hackathon Emergence Research Lab, IISc Bangalore	04/2026
Indo-Polish Pre-Conference School on Algorithms and Combinatorics IIT Dharwad	02/2026
Discussion Meeting on Number Theory and Combinatorics Evolve Back, Coorg	12/2025
Topics in Geometric Combinatorics (NCM Workshop) Chennai Mathematical Institute	06/2025
Meru Annual Combinatorics Conference BITS Pilani, K. K. Birla Goa Campus	05/2025
Geometry, Probability, and Algorithms International Centre for Theoretical Sciences, Bengaluru	05/2025
HDXs and Codes International Centre for Theoretical Sciences, Bengaluru	04/2025
Lean for the Curious Mathematician International Centre for Theoretical Sciences, Bengaluru	04/2025
Sparse graphs: treewidth, planarity, bounded expansion (GIAN) IIT Dharwad	02/2025
Combinatorial Methods in Enumerative Algebra International Centre for Theoretical Sciences, Bengaluru	12/2024
Discrete Integrable Systems International Centre for Theoretical Sciences, Bengaluru	10/2024
Meru Annual Combinatorics Conference Graphic Era Hill University, Bhimtal Campus	06/2024
A First Second Course in the Probabilistic Method IIT Palakkad	02/2024
Conference on Algorithms and Discrete Applied Mathematics (CALDAM) Dhirubhai Ambani Institute of Information and Communication Technology, Gandhinagar	02/2023
Indo-Dutch Pre-Conference School on Algorithms and Combinatorics Dhirubhai Ambani Institute of Information and Communication Technology, Gandhinagar	02/2023
Meru Annual Combinatorics Conference Pondicherry University	05/2023